

Unified Lifelong Action Learning: A Real-Video Pipeline for Continual Learning, Distillation, and Few-Shot Meta-Adaptation on UCF101

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Abstract—We present *Unified Lifelong Action Learning* (ULAL), an end-to-end pipeline that studies how a video action recogniser can keep learning new actions over time without forgetting the old ones, trained directly on real video frames (no cached features). The pipeline (i) builds an honest, group-disjoint data split that removes the leakage in UCF101’s shipped split, (ii) selects a ResNet50+LSTM backbone from a five-way bake-off, (iii) extends it to new classes with five continual-learning strategies (naive, EWC, LwF, replay, replay+LwF), (iv) compresses the best model into a MobileNetV3-Small student with four distillation objectives, and (v) adapts the student to *new, few-shot* classes with first-order MAML, using a matched fine-tune baseline so the meta-learning effect is isolated. For each stage we report the measured results and, crucially, *explain why* they come out the way they do from the abilities of each method. A final capstone notebook (ULAL) chains the best choice of every stage into one pipeline and is evaluated on real video; it is described here and left to a final run. Our headline findings: a leakage-free split lowers but trusts the numbers; rehearsal (replay+LwF) is by far the strongest continual learner; distillation buys roughly $20\times$ compression at a small accuracy cost; and meta-learning helps specifically in the few-shot regime, where *MAML+replay* is the best-balanced adapter once accuracy is scored with a metric that does not let the many old classes drown out the few new ones.

Index Terms—continual learning, catastrophic forgetting, action recognition, knowledge distillation, meta-learning, few-shot learning.

I. INTRODUCTION

Deep video models forget. Fine-tuning a trained recogniser on new action classes collapses its accuracy on the old ones, a failure called *catastrophic forgetting* [1]. Lifelong (continual) learning aims to add classes over time while retaining what was already learned. Video makes this harder than images: clips are large and temporal, so storing and replaying past data is expensive, and the model is bigger and slower to adapt.

This paper describes a single, reproducible pipeline on UCF101 [8] that trains *directly on real video frames*—no cached embeddings—so every number reflects the model you would actually deploy. Each stage is one notebook, and each stage answers one question: *how do we measure honestly?*

which backbone? how do we stop forgetting? how do we shrink the model? and how do we learn brand-new classes from only a handful of clips? Our contributions are:

- **An honest split.** UCF101’s shipped split mixes clips of the same source video across train and test; we quantify the leakage and replace it with a group-disjoint split, giving lower but trustworthy numbers.
- **A staged, controlled comparison** of five continual-learning methods, four distillation objectives, and a nine-arm meta-learning study whose fine-tune baseline isolates the meta-learning effect.
- **An explanation, not just a table.** For every stage we tie the result to the *ability* of each method, so the ordering of the numbers is understandable rather than reported.

II. RELATED WORK

Continual learning. Regularisation methods such as EWC [1] penalise changes to weights important for old tasks. Distillation methods such as LwF [2] keep old-class outputs stable using the model’s own past predictions. Rehearsal stores and replays a few exemplars; iCaRL [3] combines replay with distillation. We evaluate all of these on video.

Knowledge distillation. A compact student can be trained to match a larger teacher’s softened logits [4] or its intermediate features. We compare logit and feature (cosine, MSE) objectives, following the structure of the standard distillation tutorial.

Meta- and few-shot learning. MAML [5] learns an initialisation that adapts quickly from few examples; it is the basis of our stage-5 study. Because the base classes greatly outnumber the new ones, we report the harmonic mean of base and new accuracy, the standard metric in few-shot class-incremental learning.

III. METHOD AND MODEL

The recogniser is a CNN backbone applied to each of the 16 frames of a clip, an LSTM over those frame features, and a linear classifier on the final LSTM state. The classifier *grows*

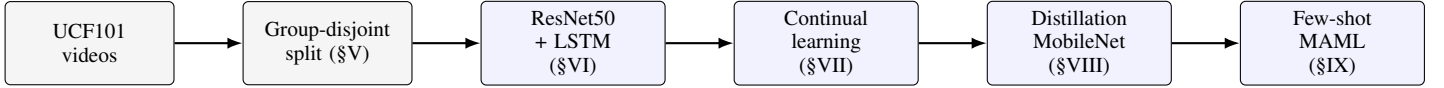


Fig. 1. The ULAL pipeline. Video is split by source group (no leakage); a ResNet50+LSTM teacher is trained on the base classes, extended to new classes by continual learning, compressed into a MobileNet student by distillation, and finally adapted to brand-new few-shot classes with MAML. The capstone notebook (§X) chains the best choice from each stage.

as new classes arrive (new rows are appended, old rows kept), and only the last block of the backbone is fine-tuned, which keeps the number of trainable parameters modest. Clips are 16 frames at 224×224 , ImageNet normalised. Figure 1 shows the five stages, each described below with its methods, its measured results, and an explanation of *why* the results look as they do.

IV. EXPERIMENTAL SETUP

All stages train on 16-frame clips at 224×224 , ImageNet-normalised, on a single GPU (Google Colab T4). The group-disjoint split uses 18/3/4 groups per class for train/val/test. Each pretrained backbone freezes all but its last block; the LSTM hidden size is 256, dropout 0.4. Stage 3 uses a $10 \rightarrow 15 \rightarrow 20$ class stream (base 10, then two 5-class tasks); Stage 4 distils the resulting 20-class model; Stage 5 treats that 20-class student as the base and adds 5 new sports. Within each stage, every compared method shares the settings in Table I, so any difference in the results comes only from the mechanism under test—this is what makes each comparison fair. Seeds are fixed and reset before every training call.

TABLE I
MAIN HYPERPARAMETERS PER STAGE. SHARED WITHIN A STAGE ACROSS ALL COMPARED METHODS.

Stage	Key settings
Backbone (2)	Adam, lr 10^{-4} , 5 epochs
Continual (3)	Adam, lr 10^{-4} , 5 epochs/task; replay 15 clips/class; LwF weight 3.0, $T=5$; EWC $\lambda=2800$
Distillation (4)	Adam, lr 10^{-4} , 15 epochs; $T=2$, CE/KD 0.75/0.25
Meta-train (5)	5-way 3-shot, 60 episodes $\times$ 8 epochs, inner SGD 5 steps @ 0.05, meta-lr 10^{-4}
Adaptation (5)	8 clips/class, 5 epochs, lr 3×10^{-4} ; replay 8 clips/class; LwF 0.5, $T=2$

A. How we read the numbers

Five simple metrics recur in the paper. *Test accuracy* is the fraction of clips classified correctly on the unseen (group-disjoint) test set. *Average accuracy* (AA) is the mean accuracy over all classes seen so far, the headline number for a continual learner. *Forgetting* is how much a past task’s accuracy has dropped from its best value—high forgetting means the model overwrote old knowledge. For the few-shot stage, where 20 old classes outnumber 5 new ones, a plain average is dominated by the old classes, so we add the *balanced* mean $(\text{acc}_{\text{new}} + \text{acc}_{\text{base}})/2$ and the *harmonic* mean, which both weight new and old equally and, in the harmonic case, punish sacrificing either side. Reporting these makes a method that learns new

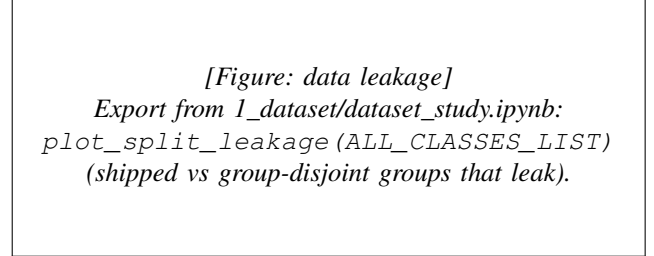


Fig. 2. Groups (source videos) that leak across train/val/test in the shipped split versus the group-disjoint split. Leakage in the shipped split inflates all downstream numbers.

classes visible, instead of one that simply guards the many old ones winning by default.

V. STAGE 1 — DATA SPLIT AND LEAKAGE

Method. A UCF101 file `v_Class_gXX_cYY.avi` encodes a *group* gXX (one source video) and a *clip* cYY (one segment of it). All clips of one group are near-duplicate frames. The shipped split is made per clip, so segments of one recording land in both train and test. We measure the fraction of groups that span more than one split, then rebuild a *group-disjoint* split that assigns each whole group to a single split (18/3/4 groups per class for train/val/test).

Result. A large fraction of groups leak across splits in the shipped data; the group-disjoint split removes the leak entirely (Fig. 2). The practical effect is that every downstream accuracy *drops*, because the model can no longer be tested on footage it trained on.

Why. A model can memorise a scene rather than the action. If a test clip comes from a video the model already saw in training, it recognises the *scene* and scores high for the wrong reason. Splitting by whole group makes the test set genuinely unseen, so the numbers become honest—and lower. Every result in the rest of the paper is read against this clean split.

VI. STAGE 2 — BACKBONE SELECTION

Methods. Five per-frame backbones, each with an LSTM head, are trained on the base classes and compared on accuracy and size: a from-scratch CNN, ResNet18, ResNet50, DenseNet121, and VGG19-BN. All but the last block of each pretrained backbone is frozen.

Result. Table II shows the trade-off. The from-scratch CNN is far behind; all ImageNet-pretrained backbones are strong; ResNet50 is chosen as the teacher for its balance of accuracy, a well-understood feature space, and manageable size.

Why. The from-scratch CNN has no prior knowledge, so with a few thousand clips it cannot learn good spatial

TABLE II
BACKBONE COMPARISON ON THE BASE CLASSES (GROUP-DISJOINT TEST).

Backbone	Test acc. (%)	Params (M)
ScratchCNN	59.0	8.5
ResNet18	89.6	11.7
ResNet50 (teacher)	92.9	25.9
DenseNet121	94.3	8.5
VGG19-BN	93.9	20.4

features—hence 59%. The pretrained backbones already carry rich ImageNet features, so the LSTM only has to learn temporal combination, and they cluster in the low-90s. DenseNet is marginally most accurate at fewer parameters, but ResNet50 gives a clean, widely-used 2048-d feature that the later distillation and meta stages build on; the small accuracy gap is a deliberate trade for a simpler downstream pipeline.

VII. STAGE 3 — CONTINUAL LEARNING

Methods. Starting from the base teacher, the head is expanded for each task and the model is trained on a $10 \rightarrow 15 \rightarrow 20$ class stream with one of five strategies. *All arms share the same optimizer, learning rate, epochs, and seed; only the mechanism (and its own strength knob) differs, so the comparison is fair.*

- **Naive:** plain fine-tuning; the lower bound.
- **EWC** [1]: a Fisher-weighted penalty holds important weights near their old values; stores no data.
- **LwF** [2]: a frozen pre-task copy supplies soft targets for the old-class logits; stores no data.
- **Replay:** a small buffer of past clips (15 per class) is mixed into every batch.
- **Replay+LwF** [3]: rehearsal and distillation together.

Result. Table III reports average accuracy (AA) over all classes seen and mean forgetting after the full stream, against a joint-training ceiling. Figure 3 shows each task’s accuracy across the whole stream.

TABLE III
CONTINUAL LEARNING AFTER THE $10 \rightarrow 15 \rightarrow 20$ STREAM. AA: AVERAGE ACCURACY; FORGET: MEAN DROP ON OLD TASKS; MEM: STORED CLIPS PER CLASS.

Method	AA (%)	Forget (%)	Mem
Naive	29.2	91.1	0
EWC	35.2	81.9	0
LwF	44.0	68.1	0
Replay	70.4	26.8	15
Replay+LwF	77.4	13.1	15
Joint (ceiling)	80.2	–	all

Why (per ability). The ordering follows exactly from what each method can do. *Naive* has no memory ability at all: the shared weights are freely overwritten to fit the new task, so old-class accuracy is destroyed (forgetting 91%). *EWC* can only slow weight change through a penalty; because video features shift a lot between action groups, the penalty is

[Figure: forgetting curve]
Export from a 3_continual_learning notebook (e.g.
replay_lwf.ipynb):
plot_forgetting_history(forget_hist,
cfg.CL_EPOCHS).

Fig. 3. Per-task test accuracy across the whole stream. Naive’s old-task lines collapse when a new task is learned; replay+LwF holds every line high.

not enough to protect the old decision boundaries, giving a modest gain. *LwF* keeps the old-class *outputs* stable using the model’s own past predictions, which is a stronger signal than a weight penalty, but with no old data it still drifts. *Replay* re-exposes the model to actual old clips, directly refreshing the old classes—this is why it jumps to 70%. *Replay+LwF* adds output-stability on top of re-exposure, combining the two abilities, and reaches 77%, within three points of the joint ceiling. The lesson is that *storing a little data beats any data-free regulariser* on video.

VIII. STAGE 4 — KNOWLEDGE DISTILLATION

Methods. The best continual model (replay+LwF, a ResNet50 teacher) is distilled into a MobileNetV3-Small+LSTM student trained on all classes. We compare four objectives from the standard distillation tutorial: cross-entropy only (CE), CE+KD (softened logits [4]), and CE with a cosine or a regressor-MSE loss between the 256-d LSTM states of teacher and student.

Result. Table IV shows the student recovers most of the teacher’s accuracy at roughly $20\times$ fewer parameters. The four objectives are close; soft-target KD gives a small edge over CE alone.

TABLE IV
DISTILLATION INTO THE MOBILENETV3-SMALL STUDENT (20-CLASS TEST).

Model	Test acc. (%)	Params (M)
Teacher (replay+LwF, ResNet50)	77.4	25.9
Student, CE only	71.5	≈ 1.5
Student, CE + KD	72.5	≈ 1.5
Student, CE + cosine	71.5	≈ 1.5
Student, CE + MSE (regressor)	72.4	≈ 1.5

Why. The compression works because the teacher’s decision function transfers: the student, guided by the teacher, reaches teacher-level behaviour with a fraction of the weights. The reason KD only *slightly* beats CE is the student backbone is itself ImageNet-pretrained and therefore already strong; the labels alone get it most of the way, so the extra “dark knowledge” in the soft targets adds only a small amount. The feature objectives (cosine, MSE) are similar because they push the student’s 256-d state toward the teacher’s, which helps a little but is constrained by the different backbones.

The honest headline is therefore *compression* (teacher-level accuracy at $\sim 20\times$ fewer parameters), not “KD $\gg$ CE”, which only appears when the student is weak or data is scarce.

IX. STAGE 5 — FEW-SHOT META-ADAPTATION

Setup and methods. Stage 5 asks a different question: given the deployed student that already knows 20 classes, how well can it learn *brand-new* classes from only a handful of clips? We keep 20 base classes and add 5 new held-out sports (a 5-way task, matching the 5-way episodes used in meta-training). Every method uses the *same* few-shot adaptation—the same support clips, the same epochs, the same learning rate, the same seed, and the same replay buffer / LwF teacher—so the only differences are the *initialisation* and whether replay/LwF are on. Three initialisations are crossed with three protection settings, giving nine arms:

- **Init:** *plain* (the student as-is), *fine-tune* (the student after ordinary supervised training on 30 auxiliary sports), and *MAML* (the student after first-order MAML meta-training [5] on those same 30 sports).
- **Protection:** *none*, *replay*, or *replay+LwF*.

The *fine-tune* init is the key control: it saw exactly the same extra data as MAML but trained in the ordinary way, so comparing MAML to fine-tune isolates the meta-learning effect from the effect of merely seeing more data.

Metric. Because 20 base classes vastly outnumber 5 new ones, a plain overall accuracy is dominated by the base (20/25 of the weight) and rewards a method that simply guards the base while learning nothing new. We therefore report the *harmonic mean* of base and new accuracy (the standard few-shot class-incremental metric), which weights the two equally and punishes sacrificing either:

$$\text{harmonic} = \frac{2 \cdot \text{acc}_{\text{new}} \cdot \text{acc}_{\text{base}}}{\text{acc}_{\text{new}} + \text{acc}_{\text{base}}}. \quad (1)$$

Result. Table V reports new-class accuracy, base retention, the balanced mean, and the harmonic mean, ranked by harmonic. *MAML+replay* is the best-balanced method; the meta init lifts new-class accuracy while replay protects the base. Figures 4–5 show the per-method comparison and a readable old-vs-new confusion.

TABLE V
FEW-SHOT META-ADAPTATION: 20 BASE + 5 NEW (5-WAY), SAME FAIR ADAPTATION FOR ALL ARMS. RANKED BY HARMONIC MEAN OF BASE AND NEW ACCURACY.

Method	New	Base	Bal.	Harm.
MAML+replay	0.330	0.624	0.477	0.432
replay	0.258	0.661	0.460	0.371
fine-tune+replay	0.170	0.681	0.426	0.272
replay+LwF	0.145	0.658	0.402	0.238
MAML+replay+LwF	0.142	0.679	0.411	0.235
fine-tune+replay+LwF	0.105	0.636	0.370	0.180
no-meta (plain)	0.367	0.055	0.211	0.096
MAML	0.357	0.000	0.178	0.000
fine-tune	0.323	0.000	0.162	0.000

[Figure: per-method new vs base]
Export from 5_maml/meta_student.ipynb: the “Compare all methods” bar chart.

Fig. 4. New-class accuracy and base retention per method. No-protection inits (MAML, fine-tune, no-meta) learn new classes but wipe the base; adding replay recovers the base, and the MAML init gives the highest new-class accuracy among the protected arms.

[Figure: 2x2 old-vs-new confusion]
Export from 5_maml/meta_student.ipynb:
plot_old_new_confusion(...) for
MAML+replay.

Fig. 5. Row-normalised old-vs-new confusion for MAML+replay: most base clips stay predicted as base and most new clips are recognised as new. Without replay this matrix collapses (all mass predicted “new”).

Why (per ability). Each row of Table V follows from what the arm can do. *MAML alone* has the best new-class accuracy but its base retention is zero: a meta-learned initialisation is built to move its features quickly onto the new task, and with no protection it moves them all the way, erasing the old classes. *Replay alone* does the opposite: it protects the base by re-exposure but starts from an init that was never primed for the new sports, so its new-class accuracy is lower. *MAML+replay* combines the two abilities—a fast-adapting init plus re-exposure—and therefore tops the harmonic ranking. The *fine-tune* arms are the honest control: because they saw the same 30 sports, *fine-tune+replay* is a strong baseline, but *MAML+replay* still beats it, which shows the gain comes specifically from meta-learning and not merely from the extra data. Finally, the LwF variants over-protect: with only 5 few-shot clips, distilling the old outputs crushes the fresh new-class head, so their new-class accuracy is lowest.

When meta-learning helps. MAML’s advantage is a property of the *init*, so it is largest when adaptation is short and data is scarce (few clips, few epochs). As the number of support clips or epochs grows, ordinary fine-tuning reaches the same solution and the meta advantage shrinks (Fig. 6, left). This is why we report the few-shot regime: it is the setting where the meta init genuinely pays off. Figure 6 (right) also shows that meta-training itself is learning—the meta query accuracy rises across epochs, confirming the init is being shaped rather than left at the student’s starting point.

X. THE ULAL CAPSTONE (TO BE RUN)

The final notebook, `ULAL.ipynb`, chains the best choice of every stage into a single lifelong pipeline and is evaluated

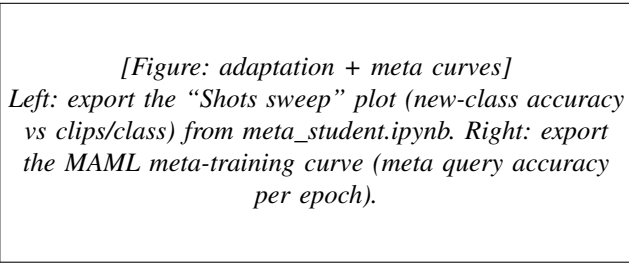


Fig. 6. Left: new-class accuracy versus the number of support clips; the MAML gap over the plain/fine-tune inits is largest at the fewest clips and narrows as data grows. Right: MAML meta-training curve—meta query accuracy rising across epochs shows the initialisation is genuinely being learned.

on real video. *This section describes the intended pipeline; the final run is pending and the numbers below are expectations, not measurements.*

What happens, stage by stage.

- 1) **Base teacher.** Train ResNet50+LSTM on the base classes—the strongest, best-understood backbone from Stage 2.
- 2) **Continual learning (replay).** Grow the model across the class stream with rehearsal, the winning ability from Stage 3, so the data-rich old classes are retained.
- 3) **Distillation.** Compress the grown teacher into the MobileNetV3-Small student (Stage 4), keeping most of the accuracy at $\sim 20\times$ fewer parameters for deployment.
- 4) **Few-shot MAML.** Meta-train the compact student on auxiliary novel classes, then adapt it to brand-new classes from a handful of clips (Stage 5), the regime where meta-learning helps.
- 5) **Real-video test.** Evaluate on real clips: for *known* classes this measures domain robustness (test the student directly); for *new* classes it measures learn-from-few (expand the head and few-shot adapt, expecting accuracy to rise from near zero after only a few examples).

Why this order. Each stage supplies an ability the next one needs: the backbone gives strong features; rehearsal gives data-rich retention; distillation gives a deployable size; and meta-learning gives fast few-shot extension. The capstone is the argument that these abilities *compose*—a single model that is accurate, small, retains old classes, and can still pick up new ones from little data. We expect the student to keep most of its known-class accuracy on real clips and to improve markedly on the new classes after few-shot adaptation, mirroring Stage 5. The measured numbers will be added once the run completes.

XI. DISCUSSION

Three themes run through the results. First, *measurement matters*: the group-disjoint split lowers every number but makes them trustworthy, and in Stage 5 the harmonic mean is what reveals MAML+replay’s win that a base-dominated average hides. Second, *data beats cleverness for retention*: on video, rehearsal (even a few clips per class) outperforms every data-free regulariser, and combining it with distillation

is best. Third, *each tool has a regime*: distillation’s win is compression, and meta-learning’s win is few-shot extension; asked outside their regime (e.g. MAML with plentiful data) they tie simpler baselines. Reading each method by its ability, rather than by a single leaderboard number, is what makes the pipeline coherent.

XII. LIMITATIONS

Several limits should be read alongside the results. (i) *One dataset and short streams*: we study UCF101 with streams up to 20 base + 5 new; longer streams and more datasets would test the conclusions further. (ii) *Rehearsal has a memory cost*: replay and replay+LwF store clips (15 per class in Stage 3, 8 per class in Stage 5), so their strong retention is not free—generative rehearsal is a natural next step to remove the buffer. (iii) *Meta-learning is regime-specific*: MAML’s advantage holds in the few-shot, few-epoch setting and disappears when data is plentiful, so it complements rather than replaces rehearsal. (iv) *Small student*: the MobileNetV3-Small student caps the achievable few-shot accuracy on new classes; a larger student would raise the absolute numbers at the cost of the compression story. (v) *The capstone’s real-video test* depends on downloadable clips and is sensitive to source quality; its numbers are pending a final run.

XIII. CONCLUSION

ULAL is a compact, honest, real-video pipeline for lifelong action recognition. A leakage-free split makes the numbers trustworthy; replay+LwF is the strongest continual learner; distillation yields a much smaller student at a small accuracy cost; and, scored fairly, MAML+replay is the best-balanced few-shot adapter, with a fine-tune control confirming the gain is from meta-learning. The capstone composes the best of each stage and is left to a final run. Future work includes longer streams, larger new-class sets, and generative rehearsal to remove stored exemplars.

AUTHOR CONTRIBUTIONS

- **First A. Author** — dataset study, leakage analysis, group-disjoint split, and backbone bake-off (Stages 1–2).
- **Second B. Author** — continual-learning methods and knowledge distillation (Stages 3–4).
- **Third C. Author** — few-shot meta-adaptation, the fine-tune control, and the ULAL capstone (Stages 5–6).

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